

## Lecture 12

# Invariant integration, characters, and Wigner–Eckart

**Objectives.** *Part I ran on a single engine: the average over the group. This lecture rebuilds that engine for compact groups — the invariant integral in place of the sum, constructed explicitly for  $U(1)$  and  $SU(2)$ , with the class integration formula  $\frac{1}{\pi} \int f(\theta) \sin^2(\theta/2) d\theta$  proved — and collects the dividends at once: every representation is unitarisable (closing Lecture 10's deferred clause), characters are orthonormal and complete, and the character formula  $\chi_j = \sin((2j+1)\theta/2)/\sin(\theta/2)$  repays, with interest, the loan Lecture 7 took for the crystal field. Tensor products then answer the question experiment asks daily — what happens when two angular momenta meet — through the Clebsch–Gordan series and coefficients; the Wigner–Eckart theorem compresses whole families of matrix elements into single numbers; and the applications cash out as dipole selection rules and the Landé  $g$ -factor.*

## 12.1 Invariant integration

Everything Part I proved flowed from one device: the average  $\frac{1}{|G|} \sum_{g \in G}$ , whose two virtues were that it exists and that translating the summation variable costs nothing. For a continuous group the sum is meaningless, but the virtues survive in integral form. The general statement is our next-to-last conceptual import.

The representation category in this lecture is continuous and finite-dimensional unless an  $L^2$  function space is named explicitly. Accordingly the averaging, Schur, character, and tensor-product theorems below state finite-dimensional hypotheses. They are compact-group analogues, not a licence to apply finite-group dimension counts to arbitrary Hilbert-space representations.

**Theorem 12.1 (Haar measure; imported).** *Every compact matrix Lie group  $G$  carries a unique Borel probability measure  $dg$  that is left-invariant:  $\int_G f(hg) dg = \int_G f(g) dg$  for all  $h \in G$  and continuous  $f$ . This measure is automatically right-invariant and invariant under  $g \mapsto g^{-1}$ .*

*Reference for the omitted proof:* Hall, Ch. 4, Theorem 4.28, and Appendix D on Haar measure.

We import the theorem for culture and honesty, but we will never integrate over a group whose measure we have not built by hand. For  $U(1)$  the measure is  $d\theta/2\pi$  on  $[0, 2\pi)$ , and invariance is the translation invariance of Lebesgue measure plus periodicity — two lines, both the reader's. For  $SU(2)$  the construction is geometry.

**Proposition 12.2 (The Haar measure of  $SU(2)$ ).** *Identify  $SU(2)$  with  $S^3$  by  $U(x) = x_0\mathbb{1} - i\vec{x} \cdot \vec{\sigma}$ ,*

$x \in S^3 \subset \mathbb{R}^4$  (Lecture 10's quaternion picture), and for continuous  $f$  set

$$\int_{\mathrm{SU}(2)} f(U) \, dU := \frac{1}{2\pi^2} \int_{S^3} f(U(x)) \, dS(x),$$

the normalised surface integral ( $\mathrm{vol} S^3 = 2\pi^2$ ). This is a probability measure invariant under left translation, right translation, and inversion.

*Proof.* The coordinates obey  $\sum_{\mu} x_{\mu}^2 = \frac{1}{2} \mathrm{tr}(U(x)^{\dagger} U(x))$ , a direct Pauli computation. Fix  $V \in \mathrm{SU}(2)$ . The quaternion subspace  $\mathrm{span}_{\mathbb{R}}\{\mathbb{I}, -i\sigma_a\}$  is closed under matrix products (Lecture 10), so  $w \mapsto Vw$  maps it to itself; the map is linear in the coordinates  $x$ , and it preserves the quadratic form above because  $\mathrm{tr}((Vw)^{\dagger}(Vw)) = \mathrm{tr}(w^{\dagger}V^{\dagger}Vw) = \mathrm{tr}(w^{\dagger}w)$ . Left translation is therefore an *orthogonal* linear map of  $\mathbb{R}^4$ ; it preserves  $S^3$  and its surface measure (orthogonal maps preserve Lebesgue measure, hence the shell limits defining the surface integral). The same argument word for word handles right translation  $w \mapsto wV$ , and inversion is  $U \mapsto U^{\dagger}$ , i.e.  $x \mapsto (x_0, -\vec{x})$ , orthogonal again. ■

The first dividend settles Lecture 10's largest outstanding debt.

**Theorem 12.3 (Unitarisability, and Maschke redux).** *Let  $G$  be a compact matrix Lie group with an invariant integral (for us:  $\mathrm{U}(1)$ ,  $\mathrm{SU}(2)$ , and  $\mathrm{SO}(3)$  via Sheet 12). Then:*

- (i) *every finite-dimensional continuous representation  $\Pi$  of  $G$  admits an invariant inner product, namely*

$$\langle v, w \rangle_G = \int_G \langle \Pi(g)v, \Pi(g)w \rangle \, dg,$$

*and is therefore equivalent to a unitary representation;*

- (ii) *every finite-dimensional continuous representation of  $G$  is a direct sum of irreducible subrepresentations.*

*In particular clause (ii) of Lecture 10's complete reducibility theorem — stated there, deferred here — is now proved.*

*Proof.* (i) Sesquilinearity is inherited from the original inner product and the linearity of the integral. The integrand is continuous on a compact group, so the integral exists. If  $v \neq 0$ , then  $\Pi(g)v \neq 0$  for every  $g$ , since  $\Pi(g)$  is invertible with inverse  $\Pi(g^{-1})$ . Thus  $g \mapsto \|\Pi(g)v\|^2$  is continuous and strictly positive on compact  $G$ , so it has a positive minimum  $m$ ; because the invariant integral is normalised,  $\langle v, v \rangle_G \geq m > 0$ . Finally, for  $h \in G$ ,  $\langle \Pi(h)v, \Pi(h)w \rangle_G = \int_G \langle \Pi(gh)v, \Pi(gh)w \rangle \, dg = \langle v, w \rangle_G$  by right invariance — Lecture 3's averaging proof with the sum retired. (ii) Combine (i) with the orthocomplement induction already proved in Lecture 10. ■

So the Gaussian inner product of Lecture 10 was not a fluke but an instance: averaging now manufactures an invariant inner product on *any* representation, and on an irreducible one the two constructions agree up to scale (Schur).

For characters one more tool is needed: integration over conjugacy classes. Here  $\mathrm{SU}(2)$  is unusually kind.

**Proposition 12.4 (The classes of  $\mathrm{SU}(2)$ ).** *Every  $U \in \mathrm{SU}(2)$  is conjugate, within  $\mathrm{SU}(2)$ , to exactly one*

$$u_{\theta} = e^{-i\theta\sigma_3/2} = \begin{pmatrix} e^{-i\theta/2} & 0 \\ 0 & e^{i\theta/2} \end{pmatrix}, \quad \theta \in [0, 2\pi],$$

so conjugacy classes are labelled by  $\text{tr} U = 2 \cos(\theta/2) \in [-2, 2]$ . Under  $\Phi(u_\theta) = R_z(\theta)$ , the parameter  $\theta$  is the geometric rotation angle.

*Proof.* The finite-dimensional spectral theorem (declared linear-algebra prerequisite, not the unbounded functional-calculus import of Lectures 9 and 13) diagonalises the unitary  $U$  by some  $W \in \text{U}(2)$ ; multiplying  $W$  by a phase puts it in  $\text{SU}(2)$  without changing the conjugation. With  $\det U = 1$  the eigenvalues are a pair  $e^{\pm i\theta/2}$ , and conjugating by  $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \in \text{SU}(2)$  swaps them, so  $\theta$  may be taken in  $[0, 2\pi]$ , where  $\cos(\theta/2)$  is injective: the label is unique, and it is the trace. ■

In the sphere picture the classes become visible:  $\text{tr} U(x) = 2x_0$ , so the conjugacy classes of  $\text{SU}(2)$  are the latitudes of  $S^3$  — the two poles  $\pm \mathbb{K}$  are the one-point classes  $\theta = 0, 2\pi$ , and the equator  $\theta = \pi$  (the traceless unitaries) is the fattest class. Integrating a class function is then a matter of weighing latitudes by their area.

**Theorem 12.5 (Class integration formula).** For every continuous class function  $f$  on  $\text{SU}(2)$ ,

$$\int_{\text{SU}(2)} f(U) \, dU = \frac{1}{\pi} \int_0^{2\pi} f(u_\theta) \sin^2 \frac{\theta}{2} \, d\theta. \quad (12.1)$$

*Proof.* In hyperspherical coordinates  $x_0 = \cos \psi$ ,  $\vec{x} = \sin \psi \, \hat{n}$  with  $\psi \in [0, \pi]$ ,  $\hat{n} \in S^2$ , the surface element is  $dS = \sin^2 \psi \, d\psi \, d\Omega$  — the four-dimensional polar Jacobian from the analysis course; the check  $\int_0^\pi \sin^2 \psi \, d\psi \cdot 4\pi = 2\pi^2$  recovers the volume. A class function depends only on the latitude,  $f(U(x)) = f(u_{2\psi})$  by Proposition 12.4, so the angular integral contributes its full  $4\pi$  and

$$\int_{\text{SU}(2)} f \, dU = \frac{4\pi}{2\pi^2} \int_0^\pi f(u_{2\psi}) \sin^2 \psi \, d\psi = \frac{1}{\pi} \int_0^{2\pi} f(u_\theta) \sin^2 \frac{\theta}{2} \, d\theta$$

after  $\theta = 2\psi$ . (Normalisation check:  $\frac{1}{\pi} \int_0^{2\pi} \sin^2(\theta/2) \, d\theta = 1$ .) ■

The weight  $\sin^2(\theta/2)$  is therefore not a formula to memorise but a picture: the relative area of the latitude sphere of radius  $\sin(\theta/2)$ . Finally,  $\text{SO}(3)$  inherits its Haar measure through the double cover — a function on  $\text{SO}(3)$  pulls back along  $\Phi$  to an even function on  $\text{SU}(2)$ , and one defines  $\int_{\text{SO}(3)} f := \int_{\text{SU}(2)} f \circ \Phi$ , invariance included; Sheet 12 supplies the bookkeeping.

## 12.2 Characters, again

The definitions of Part I are group-agnostic and port verbatim: the *character* of a finite-dimensional representation is  $\chi_\Pi(g) = \text{tr} \Pi(g)$ ; it is continuous, constant on conjugacy classes, invariant under equivalence, additive over direct sums, and satisfies  $\chi_\Pi(e) = \dim \Pi$  and, for unitary  $\Pi$  — which Theorem 12.3 lets us assume throughout —  $\chi_\Pi(g^{-1}) = \overline{\chi_\Pi(g)}$ . What Part I could not give us without an average is the orthogonality; with the integral it can.

**Theorem 12.6 (Orthogonality of characters).** Let  $\Pi, \Sigma$  be irreducible unitary representations of a compact group  $G$  with invariant integral. Then

$$\langle \chi_\Pi, \chi_\Sigma \rangle := \int_G \overline{\chi_\Pi(g)} \chi_\Sigma(g) \, dg = \begin{cases} 1 & \Pi \cong \Sigma, \\ 0 & \text{otherwise.} \end{cases}$$

*Proof.* Lecture 4’s proof, with the sum retired. For a linear map  $X$  from  $\Sigma$ ’s space to  $\Pi$ ’s, average the conjugated map:  $T_X = \int_G \Pi(g) X \Sigma(g)^{-1} dg$ . Left invariance gives  $\Pi(h) T_X \Sigma(h)^{-1} = T_X$ : an intertwiner. If  $\Pi \not\cong \Sigma$ , Schur’s lemma (Lecture 3) forces  $T_X = 0$  for every  $X$ ; choosing  $X$  to run over matrix units and using  $\Sigma(g)^{-1} = \Sigma(g)^\dagger$ , the entries read

$$\int_G \Pi(g)_{il} \overline{\Sigma(g)_{jk}} dg = 0,$$

and summing over  $i = l, j = k$  kills  $\langle \chi_\Sigma, \chi_\Pi \rangle$ . If  $\Pi = \Sigma$  (dimension  $d$ ), Schur gives  $T_X = c(X) \mathbb{1}$  with  $c(X) = \text{tr}(T_X)/d = \text{tr}(X)/d$ , since conjugation preserves traces under the integral; the same entry-reading now yields  $\int \Pi_{il} \overline{\Pi_{jk}} dg = \delta_{ij} \delta_{lk}/d$ , and the double trace gives  $\langle \chi_\Pi, \chi_\Pi \rangle = d/d = 1$ . ■

**Corollary 12.7 (The character calculus).** *For a compact group with invariant integral, with complete reducibility now proved: the multiplicity of an irreducible  $\Sigma$  in  $\Pi$  is  $\langle \chi_\Sigma, \chi_\Pi \rangle$ ; a representation is irreducible if and only if  $\langle \chi_\Pi, \chi_\Pi \rangle = 1$ ; two representations with equal characters are equivalent; and the projection onto the  $\Sigma$ -isotypic component is  $P_\Sigma = \dim \Sigma \int_G \overline{\chi_\Sigma(g)} \Pi(g) dg$ . The proofs are Part I’s, verbatim.*

For  $U(1)$  the irreducible characters are  $\chi_n(e^{i\theta}) = e^{in\theta}$ ,  $n \in \mathbb{Z}$  (Lecture 9), and orthogonality reads  $\frac{1}{2\pi} \int_0^{2\pi} e^{i(n-m)\theta} d\theta = \delta_{nm}$ : *Fourier orthogonality*. Completeness of the exponentials in  $L^2$  — the convergence theorem of the analysis course — says the characters of  $U(1)$  miss nothing. Fourier series, in other words, is the character theory of the circle, a sentence that closes the thread Lecture 4 opened when finite characters first behaved like frequencies.

**Theorem 12.8 (The characters of  $SU(2)$ ).** *On the class of  $u_\theta$ ,*

$$\chi_j(\theta) = \sum_{m=-j}^j e^{-im\theta} = \frac{\sin((2j+1)\theta/2)}{\sin(\theta/2)}, \quad \theta \in (0, 2\pi), \quad (12.2)$$

*with the limiting values  $\chi_j(0) = 2j+1$  and  $\chi_j(2\pi) = (-1)^{2j}(2j+1)$ .*

*Proof.*  $\Pi_j(u_\theta) = e^{-i\theta J_z}$  has eigenvalues  $e^{-im\theta}$ ,  $m = -j, \dots, j$ , by the classification theorem; the trace is a geometric sum:

$$\sum_{m=-j}^j e^{-im\theta} = \frac{e^{ij\theta} - e^{-i(j+1)\theta}}{1 - e^{-i\theta}} = \frac{e^{i(j+\frac{1}{2})\theta} - e^{-i(j+\frac{1}{2})\theta}}{e^{i\theta/2} - e^{-i\theta/2}},$$

after multiplying through by  $e^{i\theta/2}$ . The endpoint values are the sums directly — and  $\chi_j(2\pi) = (-1)^{2j}(2j+1)$  is  $\text{tr } \Pi_j(-\mathbb{1})$ , Lecture 10’s sign flip taking attendance. ■

Here, then, is the repayment Part I has been waiting for since the crystal field. Lecture 7 split the atomic  $d$ -shell in an octahedral environment by borrowing, on credit, the character of a rotation by  $\theta$  acting on the  $2\ell+1$  orbital states:  $\sin((\ell + \frac{1}{2})\theta)/\sin(\theta/2)$ . For integer  $j = \ell$  the representation and its character descend to  $SO(3)$  (Lecture 11),  $\theta$  is the geometric rotation angle (Proposition 12.4), and (12.2) is exactly the borrowed function: every step of that crystal-field computation is now a theorem. Lecture 11 promised that this lecture would need only the *traces* of the Wigner matrices; (12.2) is those traces, earning their keep.

Orthonormality can be checked against the class measure by hand: by (12.1), with  $\psi = \theta/2$ ,

$$\langle \chi_j, \chi_{j'} \rangle = \frac{2}{\pi} \int_0^\pi \sin((2j+1)\psi) \sin((2j'+1)\psi) d\psi = \delta_{jj'},$$

the orthogonality of sines on  $[0, \pi]$ . The measure's  $\sin^2(\theta/2)$  cancels the characters' denominators: the Weyl weight and the characters were made for each other. Better still, the same change of variables proves completeness, with no machinery beyond second-year Fourier series.

**Theorem 12.9 (Completeness of the  $SU(2)$  characters).** *Let  $L^2_{\text{cl}}(SU(2))$  be the Hilbert space of square-integrable class functions, where functions equal almost everywhere are identified. The characters  $\{\chi_j : j \in \frac{1}{2}\mathbb{Z}_{\geq 0}\}$  are an orthonormal basis of this space. In particular, every square-integrable class function is the  $L^2$ -limit of finite linear combinations of characters.*

*Proof.* The class parameter is  $\theta \in [0, 2\pi]$ , with the two endpoints representing the distinct one-point classes  $\mathbb{K}$  and  $-\mathbb{K}$ ; both have zero Weyl weight and hence do not affect  $L^2$  classes. The integration formula (12.1) extends from continuous to integrable class functions by the standard measure-extension theorem. With  $\theta = 2\psi$ , it therefore makes

$$Tf(\psi) = \sqrt{\frac{2}{\pi}} f(u_{2\psi}) \sin \psi, \quad 0 < \psi < \pi,$$

an isometry from  $L^2_{\text{cl}}(SU(2))$  to  $L^2(0, \pi)$ :

$$\|f\|_{L^2(SU(2))}^2 = \frac{2}{\pi} \int_0^\pi |f(u_{2\psi})|^2 \sin^2 \psi \, d\psi = \|Tf\|_{L^2(0, \pi)}^2.$$

It is onto: for  $h \in L^2(0, \pi)$  prescribe the class function by  $f(u_{2\psi}) = \sqrt{\pi/2} h(\psi)/\sin \psi$  for  $0 < \psi < \pi$ , assigning arbitrary endpoint values. Moreover,

$$T\chi_j(\psi) = \sqrt{\frac{2}{\pi}} \sin((2j+1)\psi).$$

We now import at this point the Fourier sine completeness theorem (Stein–Shakarchi, *Fourier Analysis*, Ch. 1, §1, and Ch. 3, Theorem 1.1):  $\{\sqrt{2/\pi} \sin(n\psi) : n = 1, 2, \dots\}$  is an orthonormal basis of  $L^2(0, \pi)$ . As  $j = 0, \frac{1}{2}, 1, \dots$ , the integer  $2j+1$  runs through every positive integer. Hence the characters map to that entire basis, proving the claim for arbitrary  $L^2$  class functions, not only continuous ones. ■

**Remark 12.10 (Peter–Weyl, starred).** What we have proved by hand for  $U(1)$  and  $SU(2)$  holds for every compact group: the matrix elements of the irreducible representations are dense in  $L^2(G)$ , and the irreducible characters are complete among class functions — the Peter–Weyl theorem, the proper generalisation of both Fourier analysis and the finite counting and group-algebra statements separated in Lecture 7 from their compact analogues. We state it for perspective and never use more of it than we have proved.

## 12.3 Tensor products and Clebsch–Gordan

Quantum systems compose by tensor product: two subsystems with spaces  $V$  and  $W$  form one system with space  $V \otimes W$ , and a symmetry acts on both parts at once,

$$(\Pi_1 \otimes \Pi_2)(g) = \Pi_1(g) \otimes \Pi_2(g), \quad \chi_{\Pi_1 \otimes \Pi_2} = \chi_{\Pi_1} \chi_{\Pi_2}$$

(the trace of a Kronecker product is the product of traces — Lecture 7 multiplied characters this way already, and the construction is unchanged). Lecture 10's  $\mathcal{H} = L^2(\mathbb{R}^3) \otimes \mathbb{C}^{2s+1}$  was an instance. What is genuinely new is the derivative.

**Proposition 12.11 (The product rule).** *The Lie algebra representation derived from  $\Pi_1 \otimes \Pi_2$  is*

$$\pi(X) = \pi_1(X) \otimes \mathbb{1} + \mathbb{1} \otimes \pi_2(X). \quad (12.3)$$

*Proof.* Write  $A(t) = \Pi_1(e^{tX})$ ,  $B(t) = \Pi_2(e^{tX})$  and expand the difference quotient bilinearly:

$$A(t) \otimes B(t) - A(0) \otimes B(0) = (A(t) - A(0)) \otimes B(t) + A(0) \otimes (B(t) - B(0));$$

divide by  $t$  and let  $t \rightarrow 0$ . It is the product rule, honestly. ■

In physics notation: the total angular momentum of a composite system is  $\vec{J} = \vec{J}_1 \otimes \mathbb{1} + \mathbb{1} \otimes \vec{J}_2$ , written  $\vec{J}_1 + \vec{J}_2$  by everyone — additivity of angular momentum is the product rule in Hermitian dress. The question of the section is how the composite multiplets decompose, and the characters answer it in four lines.

**Theorem 12.12 (The Clebsch–Gordan series).** *For all  $j_1, j_2 \in \frac{1}{2}\mathbb{Z}_{\geq 0}$ ,*

$$\Pi_{j_1} \otimes \Pi_{j_2} \cong \bigoplus_{j=|j_1-j_2|}^{j_1+j_2} \Pi_j, \quad (12.4)$$

*the sum in integer steps, each summand occurring exactly once.*

*Proof.* Take  $j_1 \geq j_2$ . On the class of  $u_\theta$ , use the sum form of  $\chi_{j_2}$  and the closed form of  $\chi_{j_1}$ :

$$\chi_{j_1} \chi_{j_2} = \sum_{m=-j_2}^{j_2} \frac{e^{-im\theta} (e^{i(j_1+\frac{1}{2})\theta} - e^{-i(j_1+\frac{1}{2})\theta})}{e^{i\theta/2} - e^{-i\theta/2}}.$$

In the second half of the numerator substitute  $m \mapsto -m$  (the sum is over a symmetric range); the terms regroup into

$$\chi_{j_1} \chi_{j_2} = \sum_{m=-j_2}^{j_2} \frac{\sin((j_1 + \frac{1}{2} - m)\theta)}{\sin(\theta/2)} = \sum_{j=|j_1-j_2|}^{j_1+j_2} \chi_j(\theta),$$

since  $j_1 + \frac{1}{2} - m$  runs over  $j + \frac{1}{2}$  for  $j = j_1 - j_2, \dots, j_1 + j_2$ . Equal characters mean equivalent representations (Corollary 12.7), and the series is read off; multiplicity one because each  $\chi_j$  appears once. As a check, the dimensions telescope:  $\sum_{j=|j_1-j_2|}^{j_1+j_2} (2j+1) = (2j_1+1)(2j_2+1)$ . Sheet 12 gives the second proof, by counting  $J_z$ -weights — same answer, different muscle. ■

The series tells us *which* multiplets appear; experiments need the change of basis. The product basis  $|j_1, m_1\rangle \otimes |j_2, m_2\rangle$  and the coupled basis  $|j, m\rangle$ ,  $j = |j_1 - j_2|, \dots, j_1 + j_2$ , are both orthonormal, and the unitary matrix between them defines the *Clebsch–Gordan coefficients*,

$$|j, m\rangle = \sum_{m_1+m_2=m} \langle j_1 m_1; j_2 m_2 | j m \rangle |j_1, m_1\rangle \otimes |j_2, m_2\rangle,$$

fixed up to phases. Lecture 10's positive ladder coefficients have already fixed the phases *within* each factor multiplet. The coupled-basis Condon–Shortley convention used here fixes the remaining relative phases as follows, with the tensor factors kept in the printed order. Construct the coupled multiplets in descending  $j$ . At each new top weight  $m = j$ , take the normalised vector annihilated by the total  $J_+$  and orthogonal to the higher- $j$  multiplets already constructed, and choose its overall sign so that the coefficient of the product ket with the largest allowed  $m_1$  is positive and real. Then define the rest of that entire multiplet

by Lecture 10’s positive total- $J_-$  ladder formula. Multiplicity one in (12.4) makes each step unique.

Thus the top state  $|j_1+j_2, j_1+j_2\rangle = |j_1, j_1\rangle \otimes |j_2, j_2\rangle$  is forced and positive; lower it to build that multiplet, take the orthogonal highest-weight direction at the next  $j$ , apply the stated sign rule, and repeat. Reversing the sign of *every* ket in one coupled multiplet gives an equivalent coupled basis and changes no representation-theoretic or physical result. Such a consistently sign-reversed answer is accepted unless a question explicitly requests the adopted Condon–Shortley phase; relative sign changes within a multiplet are not equivalent. The smallest case deserves to be done in public.

Before the algebra, the answer in one picture (Figure 12.1): laid out by weight  $m = m_1+m_2$ , the four states of  $\frac{1}{2} \otimes \frac{1}{2}$  regroup into a triplet and a singlet.

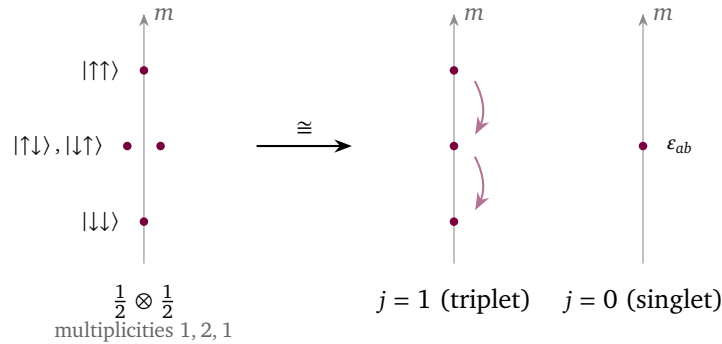


Figure 12.1: Clebsch–Gordan decomposition  $\frac{1}{2} \otimes \frac{1}{2} \cong 1 \oplus 0$ , read off the weights. The four product states distribute over  $m = m_1 + m_2$  with multiplicities 1, 2, 1 (left). The two-fold  $m = 0$  level resolves into the symmetric combination, which completes the triplet  $j = 1$ , and the antisymmetric combination  $|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle$ , the singlet  $j = 0$  — the invariant tensor  $\varepsilon_{ab}$  of  $SU(2)$ . This is the smallest instance of the general rule  $j_1 \otimes j_2 = \bigoplus_{j=|j_1-j_2|}^{j_1+j_2} j$ , and the source of the  $\varepsilon$ -tensor promised in Part I.

**Example 12.13 (Two spin-halves: singlet and triplet).** Let  $j_1 = j_2 = \frac{1}{2}$ , with product basis  $|\uparrow\uparrow\rangle, |\uparrow\downarrow\rangle, |\downarrow\uparrow\rangle, |\downarrow\downarrow\rangle$ . The series (12.4) says  $\frac{1}{2} \otimes \frac{1}{2} \cong 0 \oplus 1$ . Top state:  $|1, 1\rangle = |\uparrow\uparrow\rangle$ . Lower it: since  $J_- |\uparrow\rangle = |\downarrow\rangle$ , the product rule gives  $J_- |\uparrow\uparrow\rangle = |\downarrow\uparrow\rangle + |\uparrow\downarrow\rangle$ , while the ladder formula demands  $J_- |1, 1\rangle = \sqrt{2} |1, 0\rangle$ ; hence

$$|1, 0\rangle = \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle), \quad |1, -1\rangle = |\downarrow\downarrow\rangle \quad (\text{one more rung}),$$

and the lone direction orthogonal to  $|1, 0\rangle$  in the  $m = 0$  plane is the singlet up to sign. Here the largest allowed first-factor weight is  $m_1 = \frac{1}{2}$ , belonging to  $|\uparrow\downarrow\rangle$ , so the coupled-basis convention above selects the *singlet*

$$|0, 0\rangle = \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle), \quad J_{\pm} |0, 0\rangle = 0 \quad (\text{check it: the two terms cancel}).$$

Every Clebsch–Gordan coefficient of the problem is on display. The physics is half of low-temperature science: two spin- $\frac{1}{2}$  particles carry total spin 1 (the symmetric triplet) or 0 (the antisymmetric singlet) — the decomposition is precisely the symmetric/antisymmetric split of  $\mathbb{C}^2 \otimes \mathbb{C}^2$  under the exchange action of  $S_2$ , Part I’s smallest group commuting with  $SU(2)$ ’s biggest job, a duet that Lecture 13 will turn into a chorus. The singlet is the maximally entangled Bell state; the singlet–triplet gap organises helium’s spectrum and the hydrogen molecule’s binding.



The singlet also settles a promise about *invariant tensors*. In components,  $|0,0\rangle = \frac{1}{\sqrt{2}} \sum_{a,b} \epsilon_{ab} e_a \otimes e_b$  with  $\epsilon_{12} = -\epsilon_{21} = 1$ , and its invariance is the cofactor identity: for any  $2 \times 2$  matrix,  $A^T \epsilon A = (\det A) \epsilon$  — expand it; it is the definition of the determinant — so  $\epsilon$  is invariant under exactly the matrices of determinant one,  $SU(2)$  included. The  $\epsilon$ -tensor is the singlet wearing indices, and together with the Hermitian pairing  $\delta_{a\bar{b}}$  it generates all invariant tensors of  $SU(2)$  (stated). The analogous  $SU(3)$  classification is outside this course: the dual/conjugate and invariant-tensor language developed here is preparation for it, while Jones, Ch. 8, §§8.3–8.4, is the explicit further-reading destination.

## 12.4 Tensor operators and the Wigner–Eckart theorem

States organise into multiplets; the last step is to organise the *operators* — because what experiment measures is matrix elements  $\langle \text{final} | T | \text{initial} \rangle$  of perturbations and couplings, and these inherit the symmetry too.

**Definition 12.14 (Tensor operator).** A tensor operator of rank  $k$  on a unitary representation  $(\Pi, \mathcal{H})$  of  $SU(2)$  is a family  $T_q^k$ ,  $q = -k, \dots, k$ , of *bounded*, everywhere-defined operators  $T_q^k \in \mathcal{B}(\mathcal{H})$  transforming among themselves like the spin- $k$  multiplet:

$$\Pi(U) T_q^k \Pi(U)^{-1} = \sum_{q'} D_{q'q}^k(U) T_{q'}^k. \quad (12.5)$$

**Proposition 12.15 (Group form = algebra form).** On a finite-dimensional  $\mathcal{H}$ , (12.5) is equivalent to the bracket relations

$$[J_z, T_q^k] = q T_q^k, \quad [J_{\pm}, T_q^k] = \sqrt{k(k+1) - q(q \pm 1)} T_{q \pm 1}^k.$$

*Proof.* Conjugation  $T \mapsto \Pi(U) T \Pi(U)^{-1}$  is itself a continuous representation of  $SU(2)$  on the operator space  $\mathfrak{gl}(\mathcal{H})$ , with derivative  $T \mapsto [\pi(X), T]$  (differentiate, as for  $\text{Ad}$  in Lecture 9). Differentiating (12.5) therefore gives  $[\pi(X), T_q^k] = \sum_{q'} \pi_k(X)_{q'q} T_{q'}^k$ , and inserting the ladder basis with Lecture 10's matrix elements yields exactly the stated brackets. Conversely, the brackets say that the span of the  $T_q^k$  is an invariant subspace of  $\mathfrak{gl}(\mathcal{H})$  on which the derived representation acts by the spin- $k$  matrices; since  $SU(2)$  is connected, Lecture 9's dictionary integrates this back to (12.5). ■

The bounded hypothesis is the literal scope of the definition and theorem below. There is one standard domain-qualified extension used in physics. Let  $\mathcal{D} \subset \mathcal{H}$  be a common dense  $SU(2)$ -invariant domain, so  $\Pi(U)\mathcal{D} = \mathcal{D}$ , preserved by every  $T_q^k$  and by the derived generators. Then (12.5) may instead be read as an identity on  $\mathcal{D}$ : for  $v \in \mathcal{D}$  the conjugated product means  $\Pi(U) T_q^k \Pi(U)^{-1} v$ , and all algebra-form commutators are identities on that same  $\mathcal{D}$ . Wigner–Eckart then applies to matrix elements between finite spin multiplets contained in  $\mathcal{D}$ ; its proof uses only finite-dimensional restrictions and compressions. It does *not* make an unbounded operator everywhere defined.

Rank 0 includes a bounded operator commuting with all rotations (for example a bounded Borel function of a rotationally invariant self-adjoint Hamiltonian). Rank 1 is a *vector operator*  $\vec{V}$  repackaged in spherical components

$$V_0 = V_z, \quad V_{\pm 1} = \mp \frac{1}{\sqrt{2}} (V_x \pm iV_y),$$

which is the harmonic basis of Lecture 11's spin-one example, reappearing on schedule. Position, momentum, orbital angular momentum, and magnetic moment are generally



unbounded vector-operator examples, outside the literal bounded definition: on  $L^2(\mathbb{R}^3)$  one may take  $\mathcal{D} = \mathcal{S}(\mathbb{R}^3)$ , which rotations and the operators  $Q_i, P_i, L_i$  preserve, and read both group and commutator formulas there. Spin matrices on a fixed finite spin space are, of course, bounded. By contrast each fixed spherical harmonic  $Y_{\ell m}$  is continuous and bounded on the compact sphere, so multiplication by it on  $L^2(S^2)$  is genuinely a bounded tensor operator of rank  $\ell$ ; conjugation by rotations moves its argument, which is (12.5) for the function action.

**Theorem 12.16 (Wigner–Eckart).** *Let  $T^k$  be a bounded tensor operator in Definition 12.14. Let  $|\alpha j m\rangle$  span an invariant spin- $j$  multiplet and  $|\alpha' j' m'\rangle$  an invariant spin- $j'$  multiplet of  $(\Pi, \mathcal{H})$ , with  $\alpha, \alpha'$  any further labels. Then*

$$\langle \alpha' j' m' | T_q^k | \alpha j m \rangle = \langle j m; k q | j' m' \rangle \langle \alpha' j' || T^k || \alpha j \rangle : \quad (12.6)$$

the entire dependence on  $(m', q, m)$  is the Clebsch–Gordan coefficient, and one number — the reduced matrix element — carries all the dynamics. (Conventions differ by factors such as  $\sqrt{2j'+1}$ ; ours is fixed by (12.6).) In particular the matrix element vanishes unless

$$m' = m + q \quad \text{and} \quad |j - k| \leq j' \leq j + k.$$

*Proof.* Lecture 7’s intertwiner argument, at its sharpest. Define  $\Phi: V_j \otimes \mathbb{C}^{2k+1} \rightarrow \mathcal{H}$  by  $|j, m\rangle \otimes |k, q\rangle \mapsto T_q^k |\alpha j m\rangle$ . It intertwines: by (12.5),

$$\Pi(U) T_q^k |\alpha j m\rangle = (\Pi(U) T_q^k \Pi(U)^{-1}) \Pi(U) |\alpha j m\rangle = \sum_{q', m'} D_{q'q}^k(U) D_{m'm}^j(U) T_{q'}^k |\alpha j m'\rangle,$$

which is  $\Phi$  applied to  $(\Pi_j \otimes \Pi_k)(U)$  of the argument. Let  $P$  be the orthogonal projection onto the target multiplet — an intertwiner, since the multiplet and its complement are invariant under the unitary  $\Pi$ . Then  $P\Phi$  is an intertwiner from  $\Pi_j \otimes \Pi_k$  to  $\Pi_{j'}$ , and by the Clebsch–Gordan series (12.4) the spin  $j'$  occurs in  $\Pi_j \otimes \Pi_k$  at most once: by Schur, the space of such intertwiners is at most one-dimensional, spanned by the coupled-basis projection  $|j, m\rangle \otimes |k, q\rangle \mapsto \langle j m; k q | j' m' \rangle |j', m'\rangle$  (summed over  $m'$ ). So  $P\Phi = \lambda \cdot$  (that map), the scalar  $\lambda$  is the reduced matrix element, and taking components gives (12.6); the selection rules are the support of the Clebsch–Gordan coefficient. Where Lecture 7’s finite groups allowed multiplicities and several reduced constants,  $SU(2)$ ’s multiplicity-free series leaves exactly one. ■

The same conclusion holds for the domain-qualified extension stated after Definition 12.14, provided both displayed multiplets lie in  $\mathcal{D}$ . Indeed  $P_{j'} T_q^k|_{V_j}$  is then an ordinary map between finite-dimensional spaces, and the preceding intertwiner proof applies verbatim.

**Proposition 12.17 (Projection theorem; proof on Sheet 12).** *Within a single multiplet with  $j > 0$ , every vector operator is proportional to  $\vec{J}$ :*

$$\langle \alpha j m' | V_q | \alpha j m \rangle = \frac{\langle \vec{J} \cdot \vec{V} \rangle_{\alpha j}}{j(j+1)} \langle j m' | J_q | j m \rangle,$$

where  $\langle \vec{J} \cdot \vec{V} \rangle_{\alpha j}$  is the (constant) expectation of the scalar operator  $\vec{J} \cdot \vec{V}$  on the multiplet.

For a singlet ( $j = 0$ ), the Wigner–Eckart selection rule instead gives  $\langle \alpha 0 0 | V_q | \alpha 0 0 \rangle = 0$ : every within-multiplet matrix element of a vector operator vanishes. This is a separate statement; the quotient in the projection theorem is not defined at  $j = 0$ .

## 12.5 Applications: selection rules and the Zeeman effect

**Example 12.18 (Electric dipole selection rules).** In the dipole approximation, light couples to an atom through the position operator  $\vec{x}$  — a rank-one tensor operator, whose spherical components are proportional to  $r Y_{1q}(\hat{x})$ . A transition amplitude between orbital states is  $\langle n' \ell' m' | x_q | n \ell m \rangle$ . The bound-state multiplets used here lie in the domain of  $x_q$  and in a common rotation-invariant domain of the kind stated above. Thus this is the domain-qualified finite-multiplet extension, not a claim that  $x_q$  is bounded, and Theorem 12.16 clears the table:

$$\Delta m = q \in \{0, \pm 1\}, \quad \ell' \in \{\ell - 1, \ell, \ell + 1\},$$

which polarisation drives which  $\Delta m$  being a routine bookkeeping matter. Parity removes the middle option: under  $x \mapsto -x$  the multiplet  $\mathcal{Y}_\ell$  carries the sign  $(-1)^\ell$  — the fact Lecture 11 packed for this trip — while  $\vec{x}$  is odd, so the matrix element acquires  $(-1)^{\ell'}(-1)(-1)^\ell$  and vanishes unless  $\ell + \ell'$  is odd. Hence the textbook rules

$$\Delta \ell = \pm 1, \quad \Delta m \in \{0, \pm 1\} :$$

the same two-step logic — decompose, then intertwine — that produced Lecture 7’s molecular selection rules, now with integrals under the hood.

**Example 12.19 (Zeeman effect and the Landé  $g$ -factor).** An electron with orbital and spin angular momentum,  $\mathcal{H} = L^2(\mathbb{R}^3) \otimes \mathbb{C}^2$  and  $\vec{J} = \vec{L} + \vec{S}$  in the sense of (12.3), feels a weak field  $\vec{B} = B\hat{z}$  through  $H_Z = \mu_B B (L_z + g_s S_z)$  with  $g_s = 2$ . First-order shifts within a fixed  $(\ell, s, j)$  multiplet with  $j > 0$  are diagonal expectations, and  $L_z + 2S_z = J_z + S_z$ . Now  $\vec{S}$  is a vector operator, so the projection theorem applies:

$$\langle S_z \rangle = \frac{\langle \vec{J} \cdot \vec{S} \rangle}{j(j+1)} m, \quad \vec{J} \cdot \vec{S} = \frac{1}{2}(J^2 + S^2 - L^2) = \frac{1}{2}(j(j+1) + s(s+1) - \ell(\ell+1)),$$

the second identity from squaring  $\vec{L} = \vec{J} - \vec{S}$  and the Casimir values of Lecture 10. Hence  $\Delta E = \mu_B B g_J m$  with

$$g_J = 1 + \frac{j(j+1) + s(s+1) - \ell(\ell+1)}{2j(j+1)} : \quad (12.7)$$

the Landé  $g$ -factor, with the anomalous pattern of Zeeman multiplets falling out of one projection (full bookkeeping on Sheet 12). Each level splits into  $2j + 1$  equally spaced lines — the multiplet made visible, dimension theorem on a photographic plate.

Step back and hear the click. Everything Lecture 7 did for molecules — average, characters, decompose, intertwine, select — this lecture has done for atoms, with the single substitution  $\frac{1}{|G|} \sum \rightarrow \int_G$ . Part I’s finite machinery was not a toy version of the continuous theory; it was the theory, waiting for a better measure. Next, the machinery meets its first genuine obstruction: Lecture 13’s Heisenberg group has no faithful finite-dimensional unitary representations at all — a two-line trace argument — and quantum mechanics will force the course, at last, into infinite dimensions, where the oscillator, Fock space, and Part I’s symmetric group are waiting.

**Remark 12.20 (Computer algebra).** Notebook 12 integrates over  $SU(2)$  twice — by the parametrised class formula (12.1) and by Monte Carlo over random unitaries — and watches the answers agree; verifies character orthonormality to machine precision; rebuilds Clebsch–Gordan coefficients by the ladder algorithm and checks them against ClebschGordan and

ThreeJSymbol; and decomposes  $j_1 \otimes j_2$  by numerical character inner products — Part I’s projector code, ported from sums to integrals.

### Summary.

- *Haar measure.* Every compact group carries a bi-invariant probability measure (Theorem 12.1); on  $SU(2)$  it integrates class functions with weight  $\sin^2(\theta/2)$  (Theorem 12.5), whence unitarisability and Maschke (Theorem 12.3).
- *Orthogonality.* The  $SU(2)$  characters are orthonormal (Theorem 12.6) and complete (Theorem 12.9).
- *Characters.*  $\chi_j(\theta) = \sin((2j+1)\theta/2)/\sin(\theta/2)$  (Theorem 12.8).
- *Clebsch–Gordan.*  $j_1 \otimes j_2 = \bigoplus_{j=|j_1-j_2|}^{j_1+j_2} j$ , from the product rule for characters (Theorem 12.12, Proposition 12.11).
- *Wigner–Eckart.* A tensor operator’s matrix elements factor into a Clebsch–Gordan coefficient and one reduced matrix element (Theorem 12.16), giving selection rules and the Landé formula (eq. (12.7)).

**Tutorial.** Exercise Sheet 12.

### References

Hall Ch. 4 (compact groups, complete reducibility), Appendix C, Section C.2, Theorem C.6 (Wigner–Eckart), and the appendix on Haar measure; Tung (group integration; Wigner–Eckart); Jones Pt. 9–10; Elliott–Dawber (angular-momentum coupling); Ma (Clebsch–Gordan coefficients); Stein–Shakarchi, Ch. 3, §1, Theorem 1.1, pp. 69–79 (mean-square completeness of Fourier series, used in the character-completeness argument).

# Bibliography

The per-lecture *References* sections cite the works below by author and short title; full details are collected here. Chapter and section numbers in the lecture references follow these editions.

- M. A. Armstrong, *Groups and Symmetry*. Springer (UTM), 1988.
- V. Bargmann, On unitary ray representations of continuous groups. *Annals of Mathematics* **59** (1954), 1–46.
- A. O. Barut and R. Rączka, *Theory of Group Representations and Applications*, 2nd ed. World Scientific, 1986.
- R. Berndt, *Representations of Linear Groups: An Introduction Based on Examples from Physics and Number Theory*. Vieweg, 2007. DOI: 10.1007/978-3-8348-9401-4. Theorem 5.2, p. 71 gives the SNAG theorem used for the translation subgroup.
- D. M. Bishop, *Group Theory and Chemistry*. Dover, 1993 (orig. Oxford University Press, 1973).
- O. Bratteli and D. W. Robinson, *Operator Algebras and Quantum Statistical Mechanics 2: Equilibrium States, Models in Quantum Statistical Mechanics*, 2nd ed. (1997), 2nd printing. Springer, 2002. Section 5.2.1, pp. 7–8 gives Fock-space second quantisation, its finite-particle analytic core and closure, and the lifted unitary group.
- G. Costa and G. Fogli, *Symmetries and Group Theory in Particle Physics*. Springer (Lecture Notes in Physics 823), 2012.
- J. B. Conway, *A Course in Functional Analysis*, 1st ed. Springer (GTM 96), 1985. Chapter IX, Theorems 6.4 and 6.6 give the strong/weak bicommutant closure and the multiplication maximal-abelian theorem for sigma-finite measure spaces.
- J. P. Elliott and P. G. Dawber, *Symmetry in Physics*, 2 vols. Macmillan, 1979.
- K. Erdmann and M. J. Wildon, *Introduction to Lie Algebras*. Springer (SUMS), 2006.
- W. Fulton and J. Harris, *Representation Theory: A First Course*. Springer (GTM 129), 1991.
- G. B. Folland, *Harmonic Analysis in Phase Space*. Princeton University Press (Annals of Mathematics Studies 122), 1989. Chapter 1, Sections 2–5, including Theorem 1.50, treats the Heisenberg group, Schrödinger/Weyl representation, and Stone–von Neumann theorem.
- G. B. Folland, *A Course in Abstract Harmonic Analysis*, 2nd ed. Chapman & Hall/CRC, 2016. Section 6.6, Theorem 6.42, pp. 199–201 gives the regular semidirect-product Mackey theorem, including induced irreducibility, exhaustivity, and equivalence.
- R. Gilmore, *Lie Groups, Lie Algebras, and Some of Their Applications*. Wiley, 1974 (repr. Dover, 2005).
- B. C. Hall, *Lie Groups, Lie Algebras, and Representations: An Elementary Introduction*, 2nd ed. Springer (GTM 222), 2015.
- N. Jeevanjee, *An Introduction to Tensors and Group Theory for Physicists*, 2nd ed. Birkhäuser, 2015.
- H. F. Jones, *Groups, Representations and Physics*, 2nd ed. Institute of Physics Publishing,

1998. Chapter 8, Sections 8.3–8.4, pp. 149–164 is the course’s explicit  $SU(3)$  further-reading route; Chapter 10, Section 10.4, pp. 208–216 treats the Poincaré little groups; Appendix E, pp. 279–280 derives the covariant on-shell measure from  $\theta(p^0)\delta(p^2 - m^2)d^4p$ .
- Z.-Q. Ma, *Group Theory for Physicists*. World Scientific, 2007.
  - A. M. L. Messiah and O. W. Greenberg, Symmetrization Postulate and Its Experimental Foundation. *Physical Review* **136** (1964), B248–B267. DOI: <https://doi.org/10.1103/PhysRev.136.B248>.
  - H. Osborn, *Group Theory*. University of Cambridge, Mathematical Tripos Part III lecture notes (available from the DAMTP lecture-notes pages).
  - M. Reed and B. Simon, *Methods of Modern Mathematical Physics, Vol. I: Functional Analysis*, rev. ed. Academic Press, 1980. Chapter VIII, Theorems VIII.5, VIII.8, VIII.10, and VIII.13 and Section VIII.5, Example 2 cover spectral calculus, Stone’s theorem, invariant-domain generators, strong commutation, and the bounded-CCR obstruction.
  - P. Ramond, *Group Theory: A Physicist’s Survey*. Cambridge University Press, 2010. Chapter 11, Section 11.1.1, pp. 224–231 treats the Pauli–Lubanski Casimirs and the massless little group; pp. 230–231 distinguish finite helicity from continuous spin and record the two integer/half-odd-integer cover sectors.
  - H. Rauch, A. Zeilinger, G. Badurek, A. Wilfing, W. Bauspiess, and U. Bonse, Verification of Coherent Spinor Rotation of Fermions. *Physics Letters A* **54** (1975), 425–427. DOI: 10.1016/0375-9601(75)90798-7.
  - J. Schwichtenberg, *Physics from Symmetry*, 2nd ed. Springer, 2018.
  - J.-P. Serre, *Linear Representations of Finite Groups*. Springer (GTM 42), 1977.
  - J. Stillwell, *Naive Lie Theory*. Springer (UTM), 2008.
  - E. M. Stein and R. Shakarchi, *Fourier Analysis: An Introduction*. Princeton University Press (Princeton Lectures in Analysis I), 2003. Chapter 3, Section 1, Theorem 1.1, pp. 69–79 gives the mean-square completeness result used in Lecture 12.
  - M. Tinkham, *Group Theory and Quantum Mechanics*. McGraw–Hill, 1964 (repr. Dover, 2003).
  - W.-K. Tung, *Group Theory in Physics*. World Scientific, 1985. Chapter 10, Section 10.4, pp. 191–200 treats induced Poincaré representations, Pauli–Lubanski brackets, and massive and massless little groups.
  - S. Weinberg, *The Quantum Theory of Fields, Vol. I: Foundations*. Cambridge University Press, 1995. Section 5.6 treats causal fields and the relativistic spin–statistics connection, a theorem not derived from a rotation phase in this course.
  - S. A. Werner, R. Colella, A. W. Overhauser, and C. F. Eagen, Observation of the Phase Shift of a Neutron Due to Precession in a Magnetic Field. *Physical Review Letters* **35** (1975), 1053–1055. DOI: 10.1103/PhysRevLett.35.1053.
  - P. Woit, *Quantum Theory, Groups and Representations: An Introduction*. Springer, 2017.